INCH POUND

MIL-DTL-32209 (OS)
7 October 2008

DETAIL SPECIFICATION

CUTTER, CABLE MLU-63/B

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document gives the requirements for the manufacture, assembly, and packaging of the Cable Cutter and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	Ammunition Lot Numbering And Ammunition Data Cards
DOD-STD-2101	Classification Of Characteristics

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E11G3, Document Control, 4072 North Jackson Road, Suite 106, Indian Head, MD 20640-5115 OFFICIAL BUSINESS, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1377

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

30003 3315AS382	Cutter, Cartridge Actuated, MLU-63/B
040L3-SPI-PHST-266	Special Packaging Instruction

(Application for copies should be addressed to the DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E11G7, Document Control, 4072 North Jackson Road Suite 106, Indian Head, MD 20640-5115 OFFICIAL BUSINESS.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://e-standards.asq.org/ or from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742	Standard Practice for Radiographic Examination

(Copies of these documents are available online at http://www.astm.org or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

Rockwell Collins

431-0666-010	Wire, Antenna

(Application for copies should be addressed to Rockwell Collins, 3200 E. Renner Rd, Richardson, TX 75082.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.2.

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3.2 General. The cable cutter shall be manufactured in accordance with Drawing 30003 3315AS382

3.3 Product Baseline. The manufacturer shall maintain a product baseline list. The list shall be complete, specifying each item used in the cable cutter. The list shall contain for each item: The Description, Part Number, or Identifying Number, Revision letter and material and/or applicable specification. This list and subsequent revisions shall be submitted for technical review and shall be approved in writing by the Government Design Activity (Naval Sea Systems Command, Indian Head, MD 20640-5035). Only cable cutters conforming to this specification and approved product baseline list will be accepted for delivery.

3.4 Primary components. For the purposes of this specification, the percussion primer, ignition mix, and propellant are is defined as a primary components (see 4.3.3).

3.5 Visual Examination. Each cable cutter shall conform to Drawing 30003 3315AS382. Each Cable cutter shall be free of external damage; defects such as cracks, dents, corrosion, illegible or incomplete markings; missing or misassembled parts when inspected per 4.6.1.

3.6 Radiographic Inspection. The cable cutters shall be free of imperfections when examined per 4.6.2.

3.7 Environmental Tests.

3.7.1 Temperature-Shock/Humidity/Altitude. The cable cutters shall not function, or otherwise be adversely affected when tested per 4.6.3.1.

3.7.2 Vibration. The cable cutters shall not function, or otherwise be adversely affected when tested per 4.6.3.2.

3.8 Function. Cable cutters shall be ballistically tested per 4.6.4.

3.8.1 Function Time. Function time is the time from removal of the sear from the firing mechanism to completion of cable cut. The function time shall be less than 150 milliseconds.

3.8.2 Pull Force. The pull force to completely extract the sear from the firing mechanism shall be 15.0 pounds minimum and 35.0 pounds maximum.

3.8.3 Complete Cut. The cable cutter shall completely sever an antenna wire per Rockwell Collins Specification 431-0666-10.

3.8.4 Integrity. When tested per 4.6.4 the cable cutters shall not rupture, fragment, or vent ballistic gas.

3.9 Workmanship. Cable cutters shall be constructed and finished in a manner to assure compliance with all requirements of this specification and associated drawings. Particular attention should be directed to cracks, voids, foreign matter, missing or improperly assembled components, cracked, flaked or insufficient surface coatings, material defects, illegible or incorrect markings, corrosion, or mechanical defects such as burrs, sharp edges, and dents.

3.10 Elastomeric Materials. Elastomeric materials shall have been manufactured within 12 months of installation in the cable cutter, see 4.3.3.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.2)
b. Quality conformance inspection (see 4.5)

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of ASQ Z1.4 . Unless otherwise specified in the contract, order, or requisition, all samples for test purposes shall be provided at the expense of the contractor.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with ASQ Z1.4.

4.3.2 First article inspection. Before entering into quantity production, an acceptable first article sample of 44 cable cutters shall be prepared. This sample, manufactured using the same methods and procedures proposed for the production lot, is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract. Forty two of these cable cutters shall be expended in the tests listed in Table I and two shall be retained for investigative purposes. A first article sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of 3.5 through 3.8 of this specification. The contractor shall deliver the cable cutters to the activity designated by the contracting agency for test. First article sample acceptance shall consist of the tests shown in Table I in the order specified. Any quantity production prior to notification by the contracting agency of first article sample acceptability shall be at the contractor's risk. The first article cable cutters will not be applied as a part of the quantity specified for delivery by the contract

TABLE I First article tests.

Test/ Inspection	Requirement Paragraph	Procedure Paragraph	Group A	Group B	Group C	Group D	Total
Visual	3.5	4.6.1	18	12	12	2	44
Radiographic	3.6	4.6.2	18	12	12	2	44
TSHA	3.7.1	4.6.3.1		12			12
Vibration	3.7.2	4.6.3.2			12		12
Radiographic	3.6	4.6.2		12	12		24
Function at –65° F	3.8	4.6.4	6				6
Function at 70° F	3.8	4.6.4	6	12	12		30
Function at 160° F	3.8	4.6.4	6				6
Retained Sample						2	2

4.3.3 Production lot. Lot numbers shall be in accordance with MIL-STD-1168. Cable cutters required for all test purposes are randomly selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of cable cutters. One primary component production lot may be used in more than one cable cutter production lot. The age of elastomeric materials shall be verified by inspection of the

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manufacturers manufacture date. Cable cutters selected for the ballistic test shall be delivered to the activity designated by the contracting agency. The designated test activity shall perform the tests and examinations listed in table III to determine lot acceptance.

4.4 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of a lot inspection of all cable cutters per Table II and a sample lot acceptance per Table III. Any cable cutter which does not meet the requirements of each test and examination in Table II shall be removed from the production lot. The lot acceptance test sample shall be selected per Table IV. Lot acceptance shall be per Table III in the order shown. Any failure shall reject the lot.

TABLE II. Lot Inspections.

Item	Nature of test	Requirement	Test method	Classification (DOD-STD-2101)
Primary component	Single lot verification	3.4	4.3.3	(M101)
Elastomeric Materials	Age	3.10	4.3.3	(M102)
Cable cutter	Visual Examination	3.5	4.6.1	(M103)
Cable cutter	Radiographic Inspection	3.6	4.6.2	(M104)

TABLE III Sample Lot Acceptance Tests.

Cable cutter	Visual Examination	3.5	4.6.1	(M103)
Cable cutter	Radiographic Inspection	3.6	4.6.2	(M104)
Cable cutter	Functional Test	3.8	4.6.4	(M105)
Cable cutter	Packing, packaging, and marking	Section 5	4.5.1	(M106)

TABLE IV Lot Sample Selection

Lot Size	Sample Size	Retain
26 to 90	5	2
91 to 150	8	2
151 to 280	13	2
281 to 500	20	2
501 to 700	24	2
701 to 900	28	2
901 to 1200	32	2

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4.5.1 Packing, packaging and marking. The inspector shall ascertain that the packing and packaging of the cable cutters and the container markings conform to Drawing 040L3-SPI-PHST-266 and Section 5 of this specification, see 6.4.1.

4.6 Tests.

4.6.1 Visual Examination. Cable cutters shall be visually examined and meet the requirements of 3.5 Any cable cutter not meeting the requirements of 3.5 shall be rejected.

4.6.2 Radiographic Examination. Each cable cutter shall be radographically examined in accordance with ASTM E 1742 and meet the requirements of 3.6. Any observable imperfections in assembly shall be cause for rejection. The cable cutters shall be positioned on their sides for the most revealing exposure with the plane of the lever perpendicular to the X-ray source. All cable cutters be identified with serial numbers prior to examination. The cable cutter serial numbers shall be in consecutive order beginning with the lowest number in each production lot. The cable cutters shall be arranged on tray boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cable cutters displayed thereon. The radiograph identification shall include the complete lot number, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the X-ray. Defective cable cutters found by the vendor shall be clearly identified on the X-ray and those defectives removed from the production lot.

4.6.3 Environmental Tests.

4.6.3.1 Temperature-Shock/Humidity/Altitude. The cable cutters shall be supported on screen trays so that all areas are exposed to the prescribed atmospheric conditions at all times. The cycling scheduling is as follows:

Monday	0800	Place test items in a chamber maintained at +70°F at 50 percent relative humidity (RH).
	1200	Raise chamber temperature to +130°F and the RH to 95 percent. The chamber temperature shall reach +130°F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet (0.65 psi).
Tuesday	0800	Remove test items from above chamber and immediately place in a chamber maintained at +70°F at 50 percent RH.
	1200	Remove test items from above chamber and immediately place in chamber maintained at -65°F at pressure altitude of 70,000 feet (0.65 psi).
	1600	Remove test items from above and immediately place in a chamber maintained at 130°F at 95 percent RH.

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Wednesday	0800	Reduce chamber temperature to +70°F at 50 percent RH. The chamber temperature shall reach +70°F at 50 percent RH not later than 0900.
	1200	Raise chamber temperature to +130°F at 95 percent RH. The chamber temperature shall reach +130°F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet (0.65 psi).
Thursday	0800	Remove test items from above chamber and immediately place in a chamber maintained at +70°F at 50 percent RH.
	1200	Remove test items from the above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet (0.65 psi).
	1600	Remove test items from above chamber and immediately place in a chamber maintained at +130°F at 95 percent RH.
Friday	0800	Reduce chamber temperature to +70°F at 50 percent RH. The chamber temperature shall reach +70°F at 50 percent RH not later than 1300.
	1200	Raise chamber temperature to +130°F at 95 percent RH. The chamber temperature shall reach +130°F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -65°F at standard ambient pressure.

This schedule shall be followed for a total of 4 weeks (28 days) except that on the second and fourth weekends the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of +160°F at 95 percent RH. The cable cutters shall not fire or incur damage during the test.

4.6.3.2 Vibration. The cable cutters shall be equally divided for vibration at -65°F, 70°F, and 160°F. The cable cutters shall be mounted on a suitable fixture that allows vibration in 3 mutually perpendicular axes. The cable cutters shall be vibration cycled for one hour in each axis. The vibration cycling shall be sinusoidal vibration to the levels shown in Figure 1. The frequency shall be logarithmically swept from 5 Hz to 2000 Hz and return to 5 Hz in 20 minutes.

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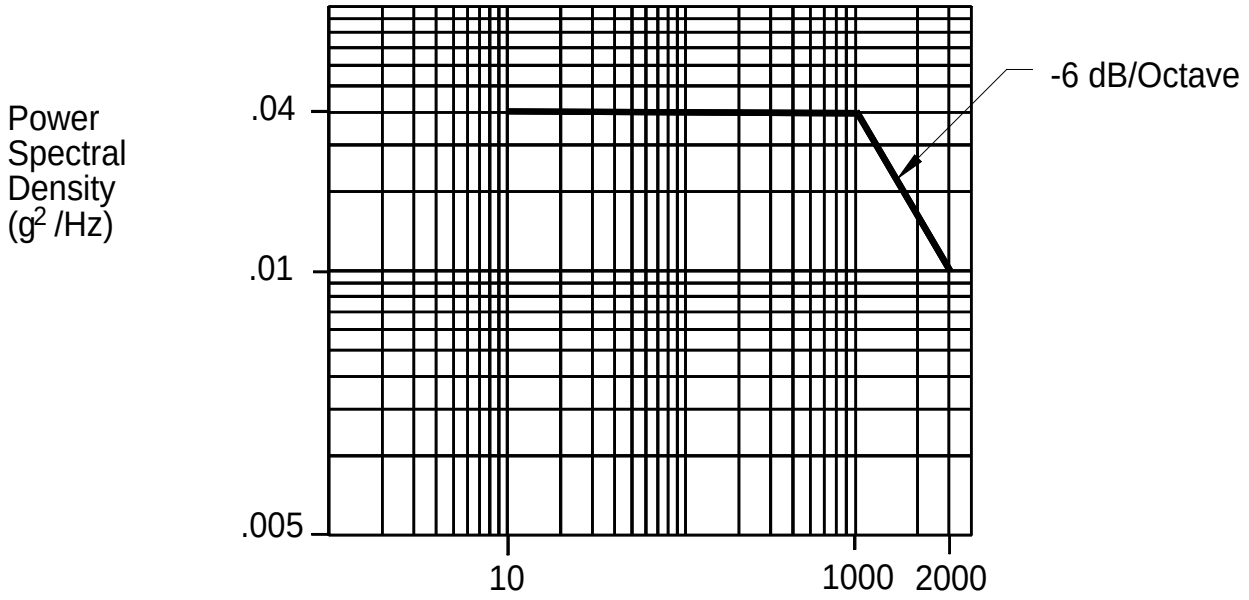


Figure 1. Vibration test levels

4.6.4 Functional Test. The cable cutters shall be equally divided for temperature conditioning at –65 ± 5°F, 70 ± 5°F, or 160 ± 5°F for a minimum of four hours. The cable cutters shall be installed in a suitable test fixture and a coaxial cable per Rockwell Collins Specification 431-0666-010 shall be inserted through the fixture. A tension of 100 pounds minimum shall then be applied to the cable. The cable cutter shall be functioned within three minutes of being removed from temperature conditioning by Applying a load of 35 pounds maximum to the firing pin sear. The cable cutters shall meet the requirements of 3.8.

4.6.4.1 Acceptance criteria. The total number of allowed defective units in a first article sample or lot acceptance sample as a result of tests and inspections of 3.5 through 3.8 shall be zero. Any defects will reject the lot represented. Screening of defects as a corrective action is at contractor's expense and the method of screening shall be subject to approval by the contracting officer.

4.6.4.2 Integrity. The functioned cable cutters shall be inspected for integrity, see 3.8.4.

4.6.4.3 Retest. There shall be no retests.

4.6.4.4 Test failure. If a test failure is attributable to an assignable cause, excluding the cable cutters, the original individual test results should be discarded and the complete test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The cable cutters are designed to sever an antenna cable in an emergency. As the cable cutters were designed for military aircraft, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).

c. Whether first article inspection is required and, if so, specify the test activity.

d. Production lot size and test activity (see 4.3.3).

e. Items of data required for each first article and production lot (see 6.3)

f. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition Ashore," OP 5 are applicable.

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government unless specified in the contract or order.

6.3.1 Standard deliverables. In addition to the delivery of Cable cutters, the contract may require delivery of the following:

a. Radiographic plates
b. Radiographic review report
c. Radiographic sample plates and techniques
d. Requests for waiver/deviation and engineering change proposal (ECP)
e. Progress and management report
f. Test and inspection report if contractor tested
g. Ammunition data cards
h. Performance oriented packaging (POP) test report
i. Process control charts.

6.4 Contract packaging and marking. The following packing and marking requirements should be specified in the contract.

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6.4.1 Preservation and packing. Level A and commercial preservation/packing should be in accordance with the Code of Federal Regulation 49 CFR 171-179. The contractor is be responsible for performance oriented packing certification and documentation in accordance with Regulation 49 CFR 171-179.

6.4.1.1 Packaging for service use. Level A preservation and packing should be used for packaging of all production lots for service use. Cable cutters should be packaged in an inner container in accordance with Drawing 040L3-SPI-PHST-266. Inner containers should be packed in the outer container in accordance with Drawing 040L3-SPI-PHST-266. For an explanation of terms see 6.4.2.

6.4.1.2 Packaging for immediate use. Commercial preservation and packing may be used for packing of all cable cutters which are intended for expenditure in first article and production lot acceptance tests. A tight pack is required. For an explanation of terms see 6.4.2.

6.4.2 Explanation of terms. For the purpose of this specification. NAVSUPINST 4030.28 (20 June 1975) paragraphs 1-4, 2-2, 203 and Appendix A, when applicable, is used to clarify terms.

6.4.3 Markings.

6.4.3.1 Special markings. Marking of exterior containers should be in accordance with 49 CFR 171-179. Markings should be applied by the contractor or his agent.

6.4.3.2 Normal markings. Unless otherwise specified in the contractor order, the marking information on unit packages and shipping containers should be as specified on Drawing 040L3-SPI-PHST-266. The specified marking information should be applied to the containers in accordance with the applicable provisions of MIL-STD-129. Lot number should be in accordance with MIL-STD-1168.

6.5 Subject term (keyword) listing.

Antenna cable
Military aircraft

Custodian
Navy – OS

Preparing activity:
Navy - OS
(Project 1377-2006-017)

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